

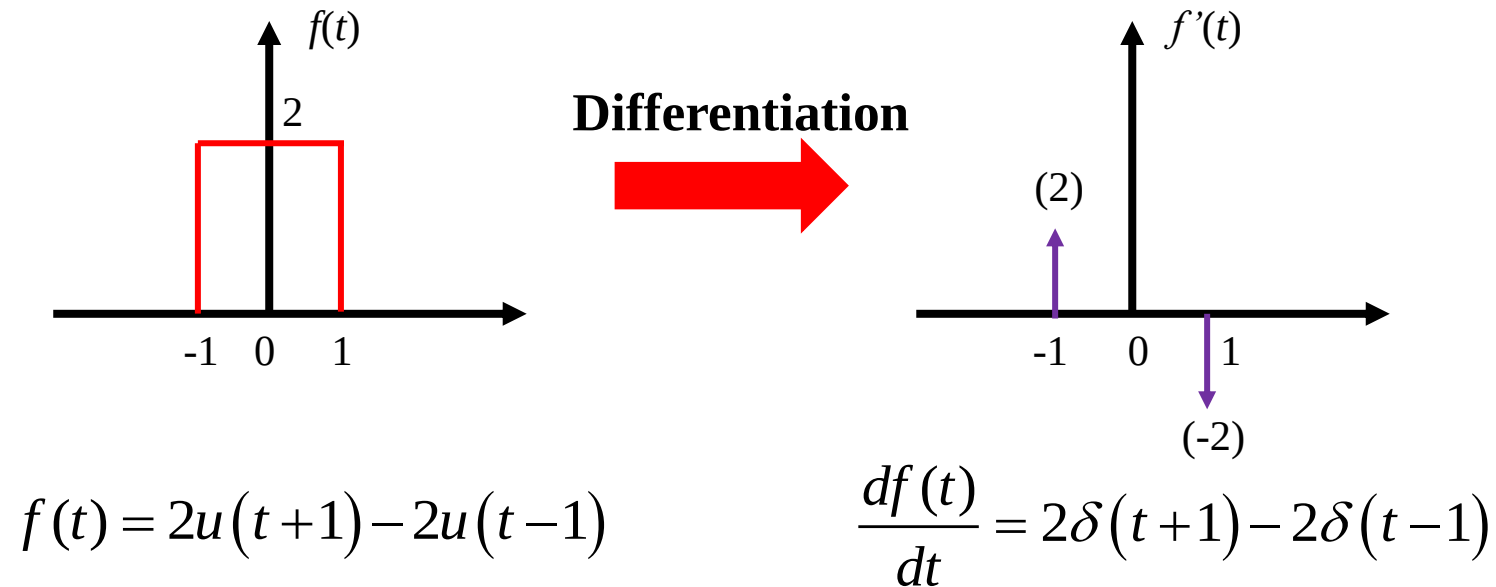
Review of Last Lecture

2.4 Jump at the Starting Point - Conversion from 0_- to 0_+ State

2.5 Zero-Input Response and Zero-State Response

2.4 Jump at the Starting Point - Conversion from 0_- to 0_+ State

Impulse Function Matching Method: Impulse functions are the cause of state jumps (for example, an impulse current in a capacitor leads to a jump in its voltage). Therefore, **at $t=0$, the $\delta(t)$ term and its derivatives of all orders on both sides of the differential equation must be balanced.**



If there is a state jump, its derivative must be an impulse function.

In general, the coefficients of the homogeneous solution can be solved by the impulse function matching method.

2.5 Zero-Input Response and Zero-State Response

1. Definition of Zero-Input Response and Determination of Undetermined Coefficients

① Definition: The response generated entirely by the initial state without the action of an external excitation signal, i.e., $r_{zi}(t) = H[\{x(0_-)\}]$

② Equation Satisfied:
$$c_0 \frac{d^n}{dt^n} r_{zi}(t) + \dots + c_{n-1} \frac{d}{dt} r_{zi}(t) + c_n r_{zi}(t) = 0$$

Therefore, it is in the form of a homogeneous solution, i.e.,

$$r_{zi}(t) = \sum_{k=1}^n A_{zik} e^{\alpha_k t}$$

where $\alpha_1, \alpha_2, \dots, \alpha_n$ are n distinct system characteristic roots.

③ Initial Conditions: $r_{zi}^{(k)}(0_+) = r_{zi}^{(k)}(0_-) = r^{(k)}(0_-)$

That is, the undetermined coefficients of the homogeneous solution $r_{zi}(t)$ can be determined by $r^{(k)}(0_-)$.

2.5 Zero-Input Response and Zero-State Response

2. Definition of Zero-State Response and Determination of Undetermined Coefficients

- ① Definition: The response generated only by the excitation when the initial state is zero.

$$r_{zs}(t) = H[e(t)]$$

- ② Equation Satisfied: $c_0 \frac{d^n}{dt^n} r_{zs}(t) + \dots + c_{n-1} \frac{d}{dt} r_{zs}(t) + c_n r_{zs}(t) = E_0 \frac{d^m}{dt^m} c(t) + \dots + E_m$

Therefore, $r_{zs}(t) = \sum_{k=1}^n A_{zsk} e^{\alpha_k t} + r_p(t)$, i.e., (It contains both the homogeneous solution and the particular solution.)

- ③ Initial Conditions: Since the k -th ($k = 0, \dots, n-1$) order derivatives of $r_{zs}(t)$ at time 0^- are all zero, i.e.,

$$r^{(k)}(0_-) = r_{zi}^{(k)}(0_-) + r_{zs}^{(k)}(0_-) = r_{zi}^{(k)}(0_-)$$

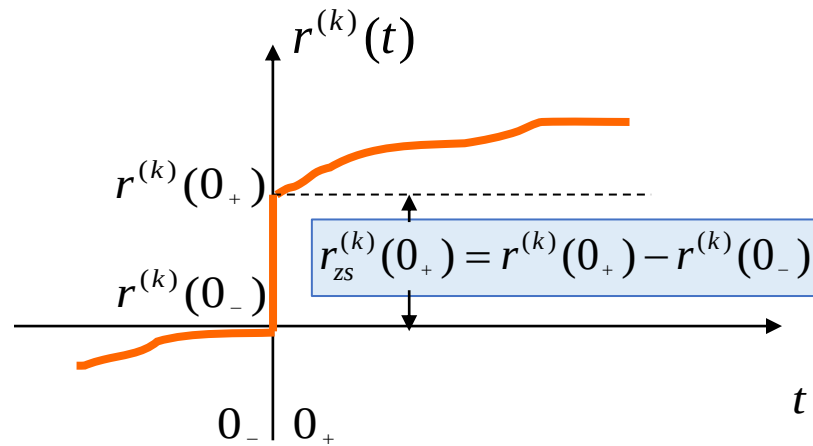
$$r_{zs}^{(k)}(0_-) = 0$$

$$r^{(k)}(0_+) = r_{zi}^{(k)}(0_+) + r_{zs}^{(k)}(0_+) = r_{zi}^{(k)}(0_-) + r_{zs}^{(k)}(0_+)$$

$$r_{zs}^{(k)}(0_+) = r^{(k)}(0_+) - r^{(k)}(0_-) = \text{Jump value}$$

Thus, A_{zsk} is determined by the jump value.

2.5 Zero-Input Response and Zero-State Response



$r^{(k)}(0_+)$: Used to determine the coefficients of the **complete response**

$r^{(k)}(0_-)$: Used to determine the coefficients of the **zero-input response**

$r^{(k)}(0_+) - r^{(k)}(0_-)$: Used to determine the coefficients of the **zero-state response**



Important!

$$\begin{aligned}
 r(t) &= \underbrace{\sum_{k=1}^n A_{Zik} e^{a_k t}}_{\text{zero-input response}} + \underbrace{\sum_{k=1}^n A_{Zsk} e^{a_k t}}_{\text{zero-state response}} + r_p(t) \\
 &= \underbrace{\sum_{k=1}^n (A_{Zik} + A_{Zsk}) e^{a_k t}}_{\text{free response}} + \underbrace{r_p(t)}_{\text{forced response}}
 \end{aligned}$$

2.5 Zero-Input Response and Zero-State Response

Key Facts

$$\text{Complete Response} = \text{Zero-Input Response} + \text{Zero-State Response}$$

Decomposing the response into zero-input response and zero-state response helps to understand the **superposition** and **homogeneity** characteristics of linear systems:

- The **zero-input response** is **linear with respect to the initial state** (initial energy storage) of the system;
- The **zero-state response** satisfies **linearity and time-invariance with respect to the excitation (input signal)** of the system.

Given a linear time-invariant system, under the same initial conditions, when the excitation is $e(t)$, the complete response is $r_1(t) = [2e^{-3t} + \sin(2t)]u(t)$. Let $r_{zi}(t)$ denote the zero-input response and $r_{zs}(t)$ the zero-state response. When the excitation is $2e(t)$, the complete response is $r_2(t) = [e^{-3t} + 2\sin(2t)]u(t)$. Which of the following is correct? ()

- ☐ A $r_{zi}(t) = 2e^{-3t}u(t), r_{zs}(t) = \sin(2t)u(t)$
- ☐ B $r_{zi}(t) = 4e^{-3t}u(t), r_{zs}(t) = [-2e^{-3t} + \sin(2t)]u(t)$
- ☐ C $r_{zi}(t) = e^{-3t}u(t), r_{zs}(t) = 2\sin(2t)u(t)$
- ☒ D $r_{zi}(t) = 3e^{-3t}u(t), r_{zs}(t) = [-e^{-3t} + \sin(2t)]u(t)$

2.5 Zero-Input Response and Zero-State Response

Example 2-7: Given a linear time-invariant system, under the same initial conditions, when the excitation is $e(t)$, the complete response is $r_1(t) = [2e^{-3t} + \sin(2t)]u(t)$; when the excitation is $2e(t)$, the complete response is $r_2(t) = [e^{-3t} + 2\sin(2t)]u(t)$.

- (1) Find the zero-input response and $r_{zi}(t)$ the zero-state response $r_{zs}(t)$.
- (2) If the initial conditions double and the excitation is $e(t-t_0)$ ($t_0 > 0$), find the complete response $r_3(t)$.

Solution: (1) We have

$$\begin{cases} r_1(t) = r_{zi}(t) + r_{zs}(t) = [2e^{-3t} + \sin(2t)]u(t) \\ r_2(t) = r_{zi}(t) + 2r_{zs}(t) = [e^{-3t} + 2\sin(2t)]u(t) \end{cases} \Rightarrow \begin{cases} r_{zi}(t) = 3e^{-3t}u(t) \\ r_{zs}(t) = [-e^{-3t} + \sin(2t)]u(t) \end{cases}$$

- (2) If the initial conditions double and the excitation is $e(t-t_0)$, the complete response is

$$\begin{aligned} r_3(t) &= 2r_{zi}(t) + r_{zs}(t-t_0) \\ &= 6e^{-3t}u(t) + [-e^{-3(t-t_0)} + \sin(2t-2t_0)]u(t-t_0) \end{aligned}$$

Contents of This Lecture

2.6 Impulse Response and Step Response

2.7 Convolution

2.8 Properties of Convolution

Learning Objectives of This Lecture

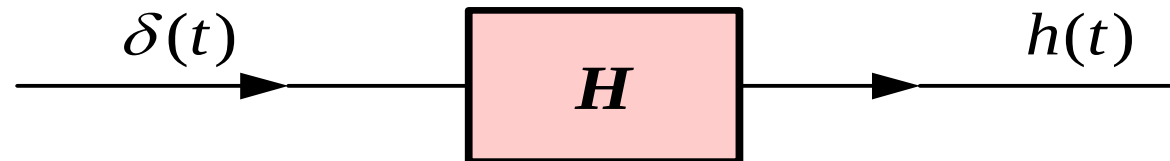
1. Skillfully apply the impulse function matching method to find the **impulse response**;
2. Be familiar with the relationship between impulse response and step response;
3. Understanding the physical meaning of **convolution**;
4. Proficiency in using the **convolutional method** to solve the **zero-state response**;

2.6 Impulse Response and Step Response

2.6.1 Impulse Response

1. Definition

The **zero-state response** generated by a system under the action of a unit impulse signal $\delta(t)$ is called the **unit impulse response** (referred to as **impulse response** for short), which is generally denoted by $h(t)$.

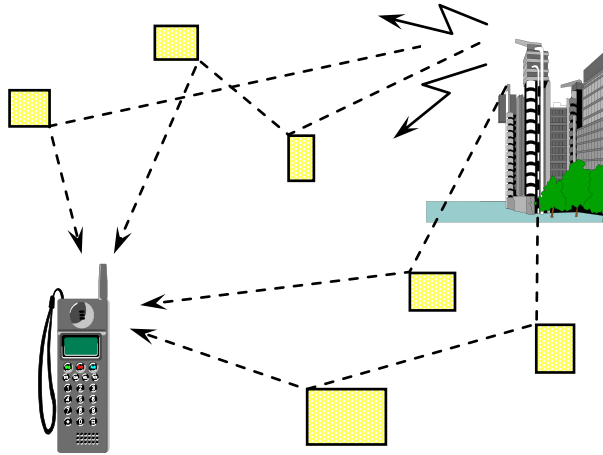


Note: In time domain, for different systems, when the same excitation $\delta(t)$ is applied under zero-state conditions, the resulting responses $h(t)$ are different. This difference indicates distinct system characteristics. Thus, **impulse response can be used to measure a system's properties.**

2.6 Impulse Response and Step Response



Think: How to characterize a wireless multipath channel using impulse response?



2.6 Impulse Response and Step Response

2. Mathematical Model of Impulse Response

An LTI system can be described by a differential equation:

$$C_0 \frac{d^n r(t)}{dt^n} + C_1 \frac{d^{n-1} r(t)}{dt^{n-1}} + \dots + C_{n-1} \frac{dr(t)}{dt} + C_n r(t) = E_0 \frac{d^m e(t)}{dt^m} + E_1 \frac{d^{m-1} e(t)}{dt^{m-1}} + \dots + E_{m-1} \frac{de(t)}{dt} + E_m e(t)$$

Response $r(t)$ and its derivatives (up to the n -th order).

Let $e(t) = \delta(t)$,
then $r(t) = h(t)$

Excitation $e(t)$ and its derivatives (up to the m -th order).

$$C_0 h^{(n)}(t) + C_1 h^{(n-1)}(t) + \dots + C_{n-1} h^{(1)}(t) + C_n h(t) = E_0 \delta^{(m)}(t) + E_1 \delta^{(m-1)}(t) + \dots + E_{m-1} \delta^{(1)}(t) + E_m \delta(t)$$

Which of the following statements are correct? ()

A

Impulse response is the **complete response** when the excitation is a unit impulse function.

B

Impulse response is the **zero-state response** when the excitation is a unit impulse function.

C

Impulse response is the **zero-input response** when the excitation is a unit impulse function.

D

An impulse input signal can be converted into the initial conditions of the system.

Submit

2.6 Impulse Response and Step Response

3. Form of the Impulse Response

$$\begin{aligned} &C_0 h^{(n)}(t) + C_1 h^{(n-1)}(t) + \cdots + C_{n-1} h^{(1)}(t) + C_n h(t) \\ &= E_0 \delta^{(m)}(t) + E_1 \delta^{(m-1)}(t) + \cdots + E_{m-1} \delta^{(1)}(t) + E_m \delta(t) \end{aligned}$$

① Related to the relative orders of n and m

If $n > m$, $h(t)$ does not contain $\delta(t)$ or its derivatives. (**most common**)

If $n = m$, $h(t)$ contains $\delta(t)$ but not its derivatives.

If $n < m$, $h(t)$ contains $\delta(t)$ and its derivatives (up to the $(m-n)$ -th order) .

2.6 Impulse Response and Step Response

3. Form of the Impulse Response

② Related to characteristic roots

If $n > m$ and the characteristic roots are simple (no repeated roots),

$$h(t) = \left[\sum_{i=1}^n A_i e^{\alpha_i t} \right] u(t)$$

For $t > 0_+$, $\delta(t)$ and its derivatives vanish (the excitation no longer exists), so the free term on the right side of the differential equation is zero. At this point:

- The impulse response is a zero-state response with the same form as the zero-input response (equivalent to solving only the homogeneous equation).
- Energy storage introduced by the impulse signal is converted into (equivalent to) the initial conditions of the zero-input response.

2.6 Impulse Response and Step Response

Example 2-8: Given the system differential equation below, find the impulse response $h(t)$.

$$\frac{d^2 r(t)}{dt^2} + 4 \frac{dr(t)}{dt} + 3r(t) = \frac{de(t)}{dt} + 2e(t)$$

Solution: Let $e(t) = \delta(t)$, $r(t) = h(t)$, we have

$$\frac{d^2 h(t)}{dt^2} + 4 \frac{dh(t)}{dt} + 3h(t) = \delta'(t) + 2\delta(t)$$

Step 1: Find characteristic roots

Characteristic equation: $\alpha^2 + 4\alpha + 3 = 0$

Characteristic roots: $\alpha_1 = -1, \alpha_2 = -3$

Step 2: Homogeneous solution form

$$h(t) = (A_1 e^{-t} + A_2 e^{-3t})u(t)$$

2.6 Impulse Response and Step Response

Step 3: Impulse Function Matching Method

$$\frac{d^2 h(t)}{dt^2} + 4 \frac{dh(t)}{dt} + 3h(t) = \delta'(t) + 2\delta(t)$$

$$\text{Let } \frac{d^2 h(t)}{dt^2} = \delta'(t) + a\delta(t), \quad \text{then } \frac{dh(t)}{dt} = \delta(t) + \textcolor{red}{a}\Delta u(t), \quad h(t) = \textcolor{red}{1} \cdot \Delta u(t)$$

Substitute them into the differential equation and balance impulses: $a + 4 = 2 \rightarrow a = -2$

Step 4: Determine initial conditions

$$h(0_+) = h(0_-) + 1 = 1, \quad h'(0_+) = h'(0_-) + a = -2$$

Step 5: Solve for the coefficients and the impulse response

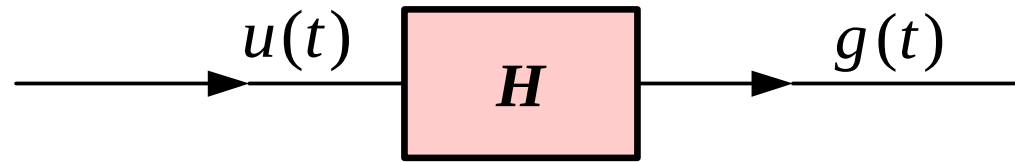
$$\begin{array}{l} h(t) = A_1 e^{-t} + A_2 e^{-3t} \\ h'(t) = -A_1 e^{-t} - 3A_2 e^{-3t} \end{array} \quad \xrightarrow{\text{Set } t=0} \quad \begin{cases} A_1 + A_2 = 1 \\ A_1 + 3A_2 = 2 \end{cases} \quad \longrightarrow \quad \begin{cases} A_1 = 1/2 \\ A_2 = 1/2 \end{cases}$$

$$\text{so } h(t) = \frac{1}{2}(e^{-t} + e^{-3t})u(t).$$

2.6 Impulse Response and Step Response

2.6.2 Step Response

The **zero-state response** generated by a system with the input of a unit step signal $u(t)$ is called the **unit step response** (or step response for short), denoted by $g(t)$.



- The right side of the system equation contains $u(t)$, so the step response includes both a homogeneous solution and a particular solution.
- For LTI systems, the step response can also be found using the relationship between impulse response and step response:

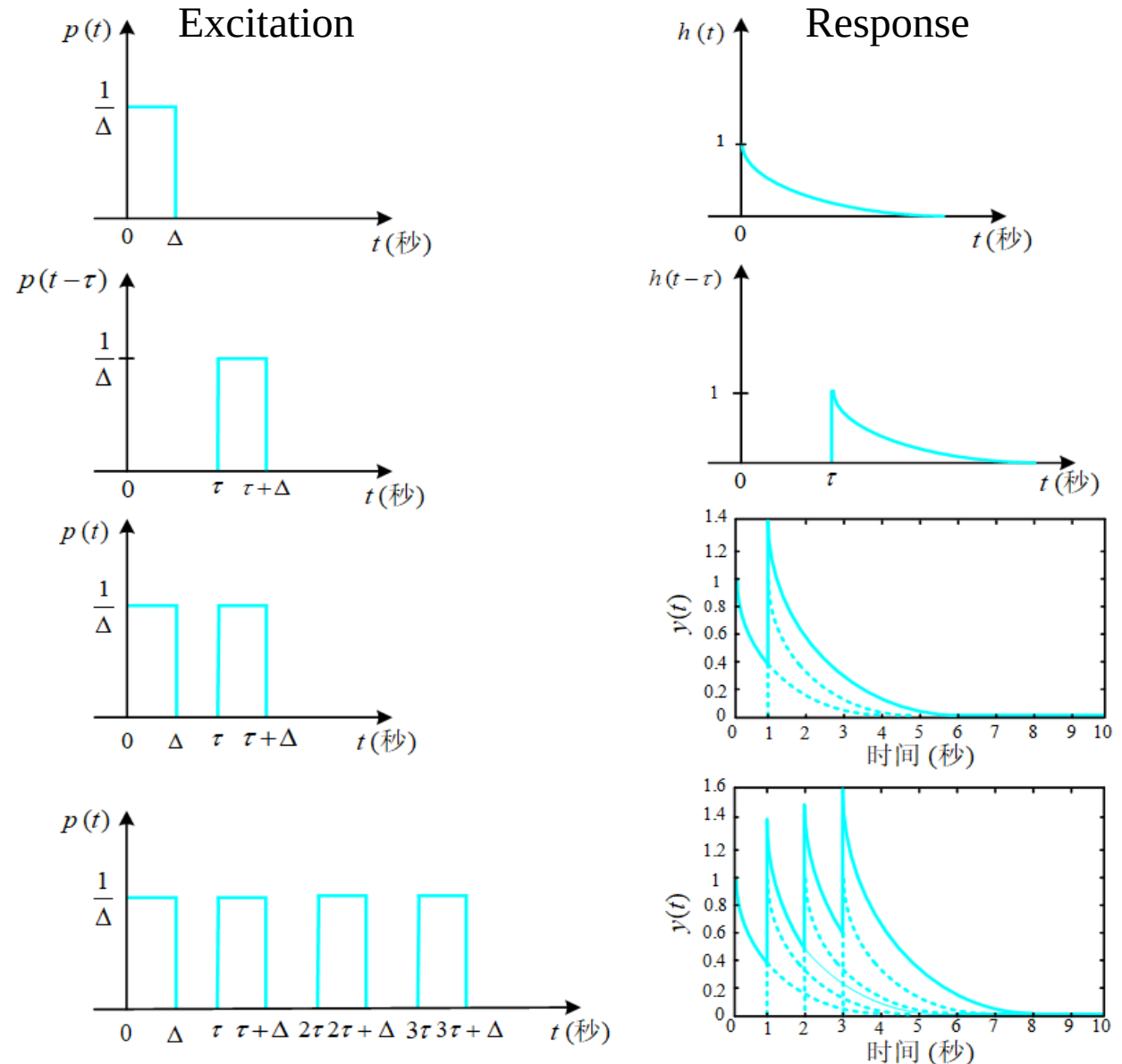
$$g(t) = \int_{0_-}^t h(\tau) d\tau \quad (\text{as } u(t) = \int_{-\infty}^t \delta(\tau) d\tau = \int_{0_-}^t \delta(\tau) d\tau)$$

- The impulse response is the derivative of the step response:

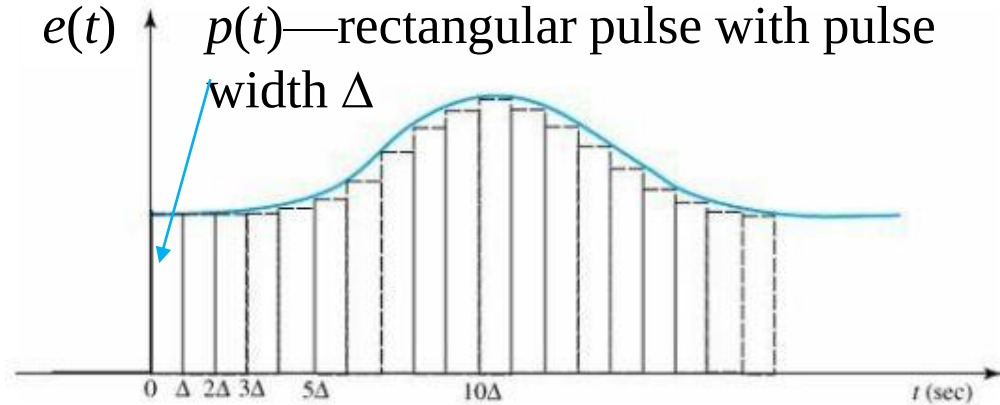
$$h(t) = \frac{dg(t)}{dt}$$

2.7 Convolution

- **Convolution** is a fundamental mathematical tool and a core technique for system analysis.
- The **Convolution Theorem** connects the time-domain analysis and transform-domain analysis of systems, running through the entire course—this makes it extremely important.



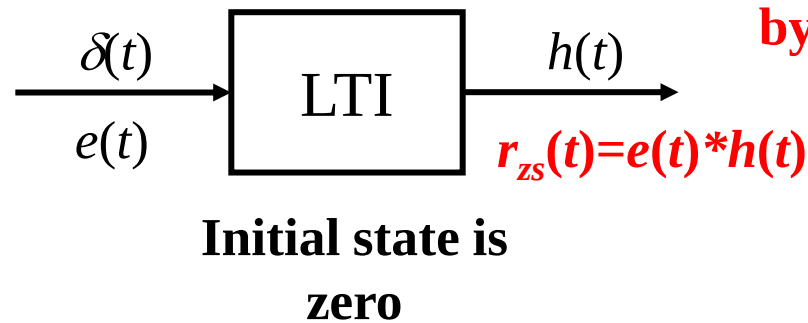
2.7 Convolution



$$e(t) = \lim_{\Delta \rightarrow 0} \sum_{i=0}^{\infty} e(i\Delta) p(t - i\Delta)$$

$$= \lim_{\Delta \rightarrow 0} \sum_{i=0}^{\infty} e(i\Delta) \frac{u(t - i\Delta) - u[t - (i+1)\Delta]}{\Delta} \Delta$$

$$e(t) = \lim_{\Delta \rightarrow 0} \sum_{i=0}^{\infty} e(i\Delta) \delta(t - i\Delta) \Delta = \int_0^{\infty} e(\tau) \delta(t - \tau) d\tau$$



by superposition, homogeneity
and time-invariance

$$r_{zs}(t) = \lim_{\Delta \rightarrow 0} \sum_{i=0}^{\infty} e(i\Delta) h(t - i\Delta) \Delta = \int_0^{\infty} e(\tau) h(t - \tau) d\tau = e(t) * h(t)$$

Convolution

The zero-state response of a system is the convolution of the excitation and the impulse response.

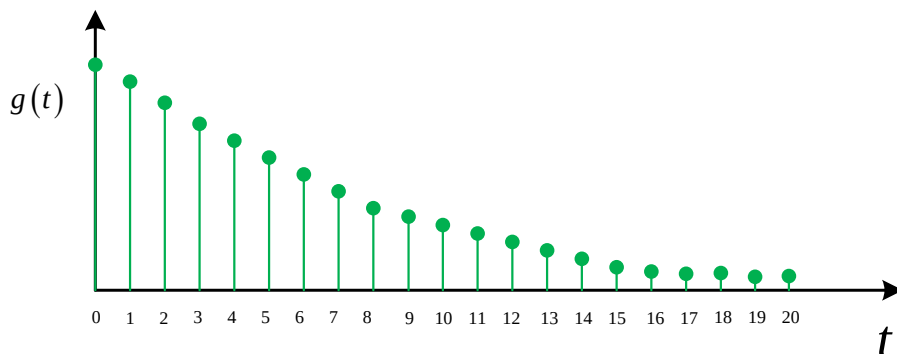
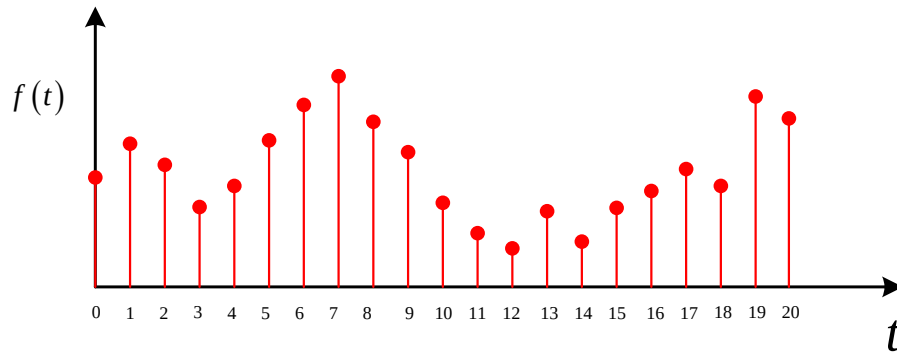
Calculation Method: Decompose any input signal into a weighted sum (integral) of impulse signals, then compute the corresponding weighted sum (integral) of the system's impulse responses.

2.7 Convolution

Definition of Convolution

For two functions $f(t)$ and $g(t)$, the integral $f(t) * g(t) = \int_{-\infty}^{\infty} f(\tau) g(t - \tau) d\tau$ is called the **convolution integral** of $f(t)$ and $g(t)$, often abbreviated to "**convolution**" and denoted by the asterisk "*".

Physical Meaning of Convolution

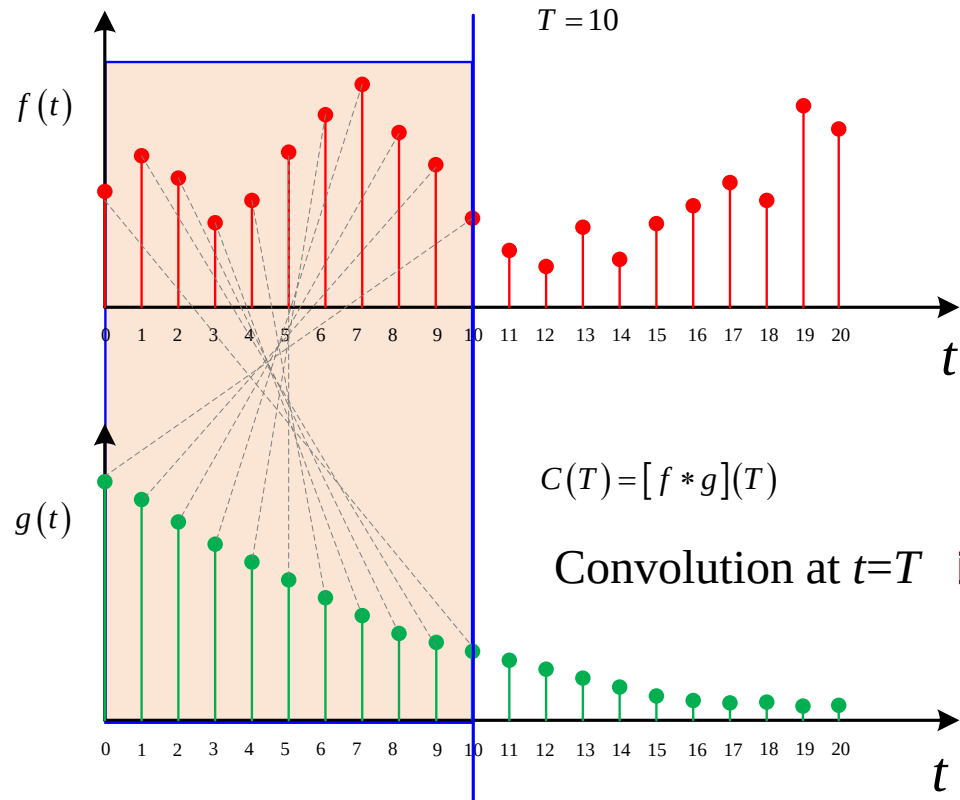


$f(t)$: Input signal of the system.

$g(t)$: Represents the attenuation trend of a signal value (treated as an impulse signal) input at a certain moment.

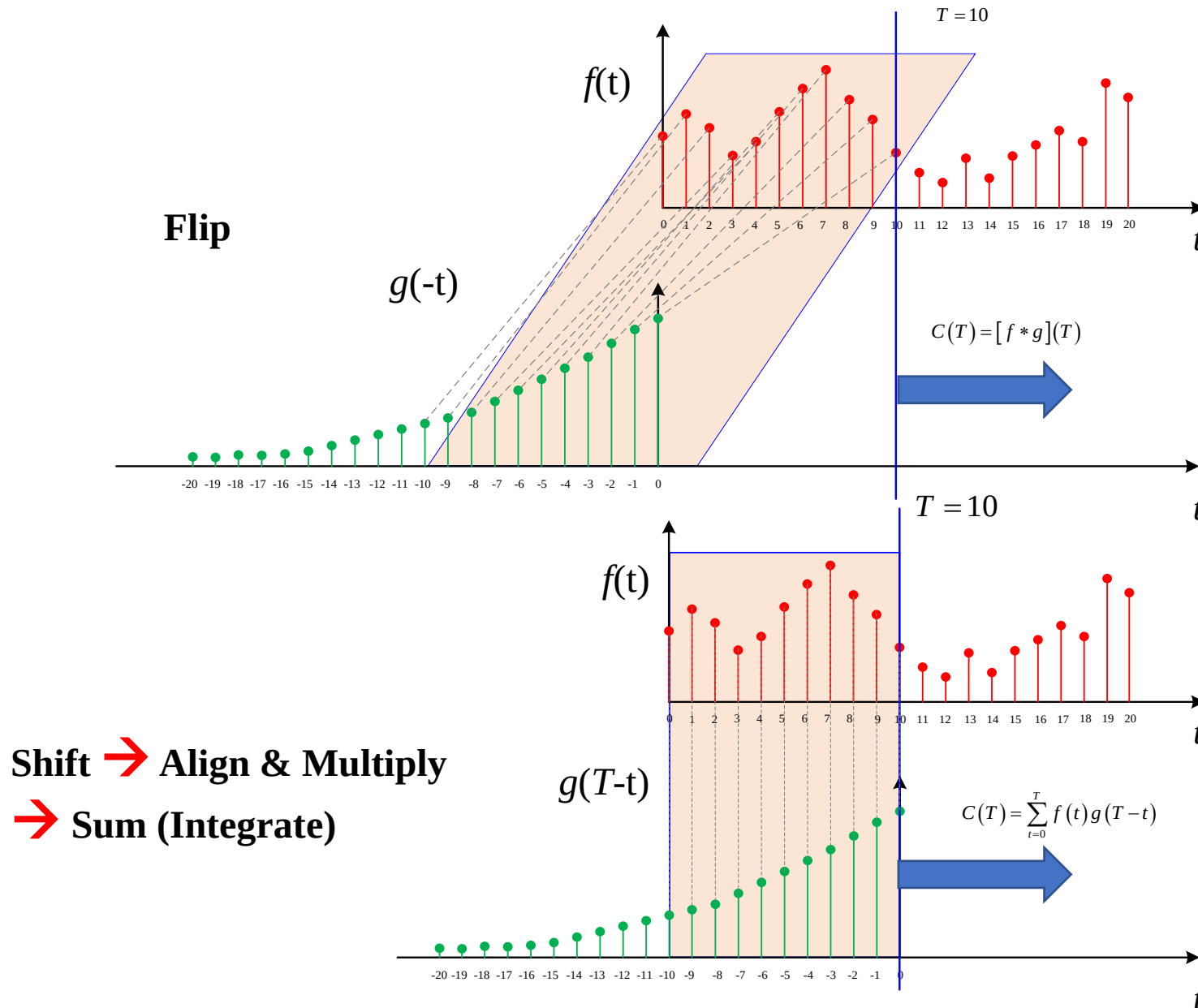
At time $t=T$, the value of the input $f(0)$ (applied at $t = 0$) attenuates to $f(0)g(T)$.

2.7 Convolution



- Since signals are input continuously, the final output of the system reflects the **cumulative effect** of all previous input signals.
- For example, at $t = 10$ (denoted as $T = 10$), the output depends on the entire shaded area in the figure.
 - $f(10)$ is just input at $t = 10$, so its contribution to the output is $f(10)g(0)$;
 - The input $f(9)$ (applied at $t = 9$) has only undergone 1 unit of time attenuation, so its output contribution is $f(9)g(1)$;
 - This pattern continues, as described by the dashed line in the original figure.
- Multiplying these corresponding pairs of values and summing the results gives the output signal value at $T = 10$ —this is exactly the convolution value of $f(t)$ and $g(t)$ at $t = 10$.

2.7 Convolution



Core Steps of Convolution (The "Four-Step Process")

1. **Flip:** Transform $g(\tau)$ to $g(-\tau)$;
2. **Shift:** Shift $g(-\tau)$ by t to get $g(t-\tau)$;
3. **Align and Multiply:** Multiply the overlapping segments of $f(\tau)$ and $g(t-\tau)$;
4. **Sum (Integrate):** Compute the integral of the product over τ to get the convolution value at time t .

2.7 Convolution

Convolution Operation Analysis for Systems

The steps to calculate the zero-state response for LTI systems by convolution:

1. Replace the independent variable t of both $e(t)$ (input) and $h(t)$ (impulse response) with τ , resulting in $e(\tau)$ and $h(\tau)$.
2. Execute the "four-step process" of convolution:

Flip: $h(\tau) \rightarrow h(-\tau)$

Shift: $h(-\tau) \rightarrow h(t - \tau) = h[-(\tau - t)]$ $\begin{cases} t < 0, & \text{Left shift} \\ t > 0, & \text{Right shift} \end{cases}$

Multiply: $e(\tau)h(t - \tau)$

Integrate: $r_{zs}(t) = e(t) * h(t) = \int_{-\infty}^{\infty} e(\tau)h(t - \tau)d\tau$

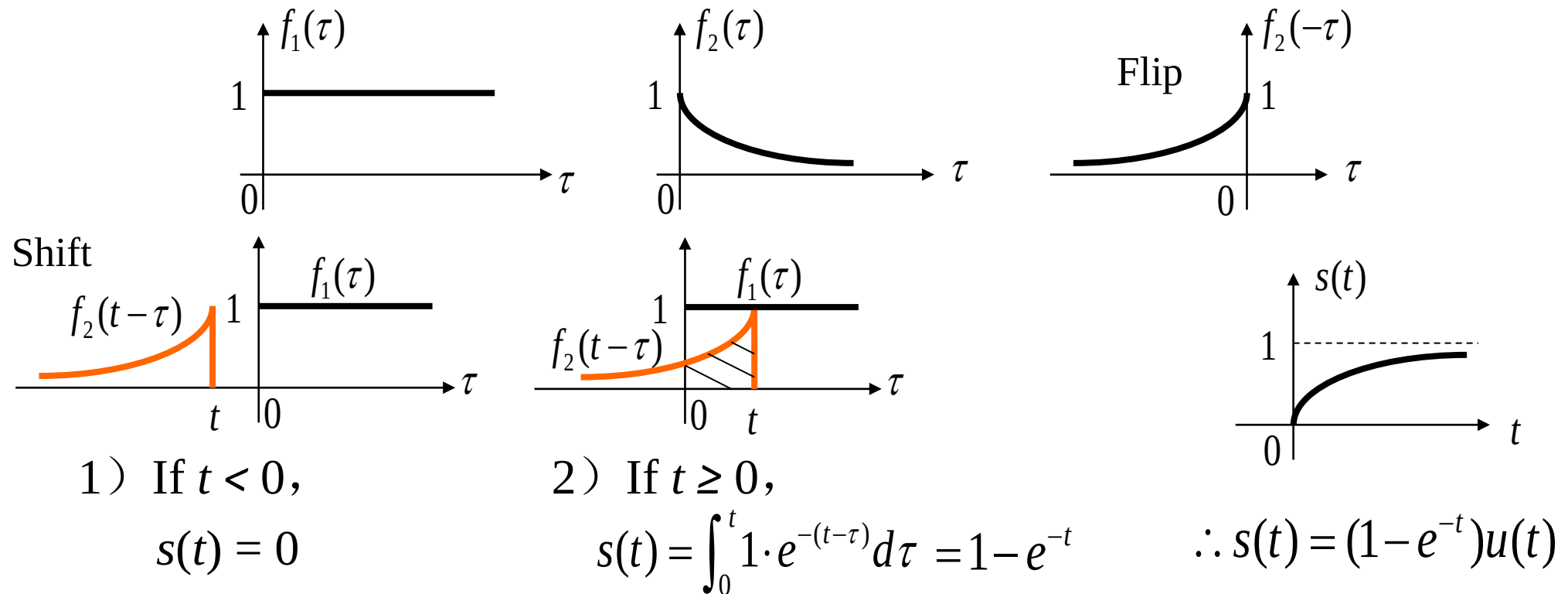
- Key Note:* The most critical part of convolution calculation is determining the **integral limits**—these limits depend on **the intervals where $e(\tau)$ and $h(t - \tau)$ overlap.**

2.7 Convolution

Example 2-9: Given $f_1(t) = u(t)$ and $f_2(t) = e^{-t}u(t)$, find their convolution $s(t) = f_1(t) * f_2(t)$

Method 1: Graphical Method

$$s(t) = \int_{-\infty}^{\infty} f_1(\tau) f_2(t - \tau) d\tau$$



2.7 Convolution

Example 2-9: Given $f_1(t) = u(t)$ and $f_2(t) = e^{-t}u(t)$, find their convolution $s(t) = f_1(t) * f_2(t)$

Method 2: Analytical Method (Direct Integration)

$$s(t) = \int_{-\infty}^{\infty} f_1(\tau) f_2(t - \tau) d\tau = \int_{-\infty}^{\infty} u(\tau) e^{-(t-\tau)} u(t - \tau) d\tau$$

Step 1: Determine the non-zero intervals of the functions

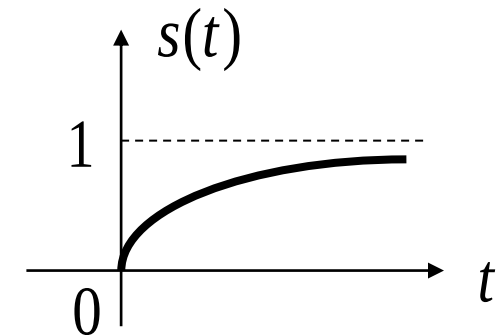
$f_1(\tau) = u(\tau)$ is non-zero only when $\tau > 0$,

$f_2(t - \tau) = e^{-(t-\tau)} u(t - \tau)$ is non-zero only when $\tau < t$.

Step 2: Determine the integral limits by the overlapping

For $t < 0$: No overlap between $\tau > 0$ and $\tau < t$.

For $t \geq 0$: Overlap interval is $0 \leq \tau \leq t$.



$$s(t) = \int_0^t e^{-(t-\tau)} d\tau = e^{-t} \int_0^t e^{\tau} d\tau = e^{-t} (e^t - 1) = 1 - e^{-t} \quad (t \geq 0)$$

Key Note: The analytical method is more efficient for signals with explicit mathematical expressions (e.g., exponential signals, step signals), while the graphical method is better for signals defined by waveforms.

2.7 Convolution

Example 2-10: Given the system's differential equation

$$\frac{d^2 r(t)}{dt^2} + 3 \frac{dr(t)}{dt} + 2r(t) = \frac{de(t)}{dt} + 3e(t)$$

It is known that the excitation $e(t) = e^{-t}u(t)$, and the impulse response is $h(t) = (2e^{-t} - e^{-2t})u(t)$, find the zero-state response.

Solution: The zero-state response of the system is

$$\begin{aligned} r_{zs}(t) &= e(t) * h(t) \\ &= 2 \int_0^t e^{-\tau} e^{-(t-\tau)} d\tau - \int_0^t e^{-\tau} e^{-2(t-\tau)} d\tau \\ &= 2e^{-t} \int_0^t d\tau - e^{-2t} \int_0^t e^{\tau} d\tau \\ &= 2te^{-t} - e^{-2t}(e^t - 1) \\ &= (2t - 1)e^{-t} + e^{-2t} \quad (t > 0) \end{aligned}$$

2.8 Properties of Convolution

Convolution has several fundamental properties that simplify system analysis. For LTI systems, these properties directly reflect the system's inherent characteristics (e.g., time-invariance, linearity).

1. Commutative Law (Exchange Law)

$$f_1(t) * f_2(t) = f_2(t) * f_1(t)$$

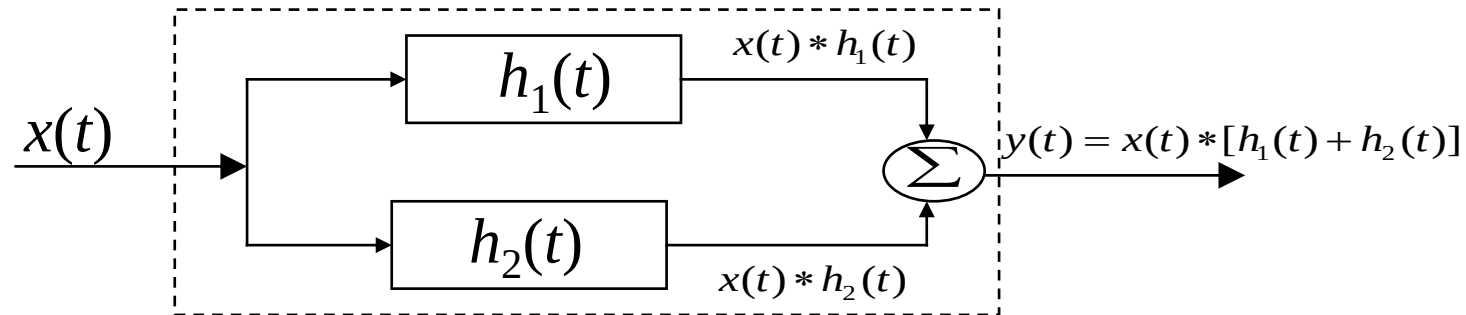
Physical Meaning: The zero-state response of a system with impulse response $h(t)$ to excitation $e(t)$ is identical to the response of a system with impulse response $e(t)$ to excitation $h(t)$.

2.8 Properties of Convolution

2. Distributive Law (Additive Law)

$$f_1(t) * [f_2(t) + f_3(t)] = f_1(t) * f_2(t) + f_1(t) * f_3(t)$$

Physical Meaning: The response of an LTI system to the sum of two excitations equals the sum of the system's responses to each individual excitation. This property directly corresponds to the **superposition principle** of LTI systems.

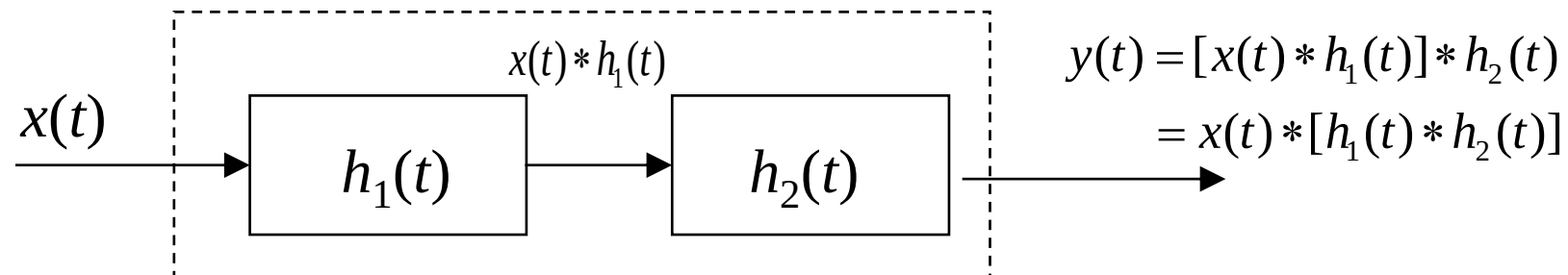


2.8 Properties of Convolution

3. Associative Law

$$[f_1(t) * f_2(t)] * f_3(t) = f_1(t) * [f_2(t) * f_3(t)]$$

Physical Meaning: Cascading two LTI systems (with impulse responses $h_1(t)$ and $h_2(t)$) is equivalent to a single LTI system with impulse response $h(t) = h_1(t) * h_2(t)$. The order of cascading does not affect the final output.



2.8 Properties of Convolution



4. Convolution with Impulse Function

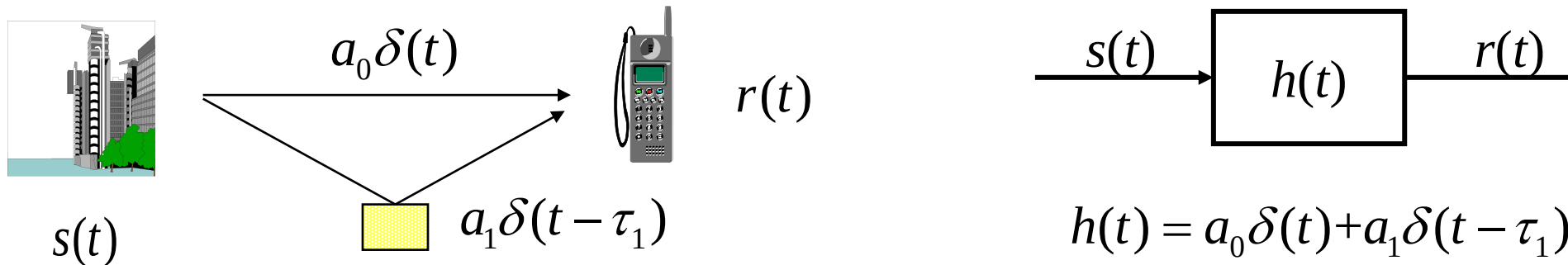
By the sampling property of the impulse function,

$$f(t) * \delta(t) = \int_{-\infty}^{\infty} f(\tau) \delta(t - \tau) d\tau = f(t)$$

$$f(t) * \delta(t - t_0) = \int_{-\infty}^{\infty} f(\tau) \delta(t - \tau - t_0) d\tau = f(t - t_0)$$

Extension: $f(t) * \delta'(t) = f'(t)$ $f(t) * u(t) = \int_{-\infty}^t f(\lambda) d\lambda$

Example: Model the received signal of a wireless communication system with a 2-path channel.



Using the properties of the **associative law** and **convolution with impulse function**, the received signal is

$$r(t) = s(t) * h(t) = s(t) * [a_0 \delta(t) + a_1 \delta(t - \tau_1)] = a_0 s(t) + a_1 s(t - \tau_1)$$

2.8 Properties of Convolution



5. Time-Shift Property

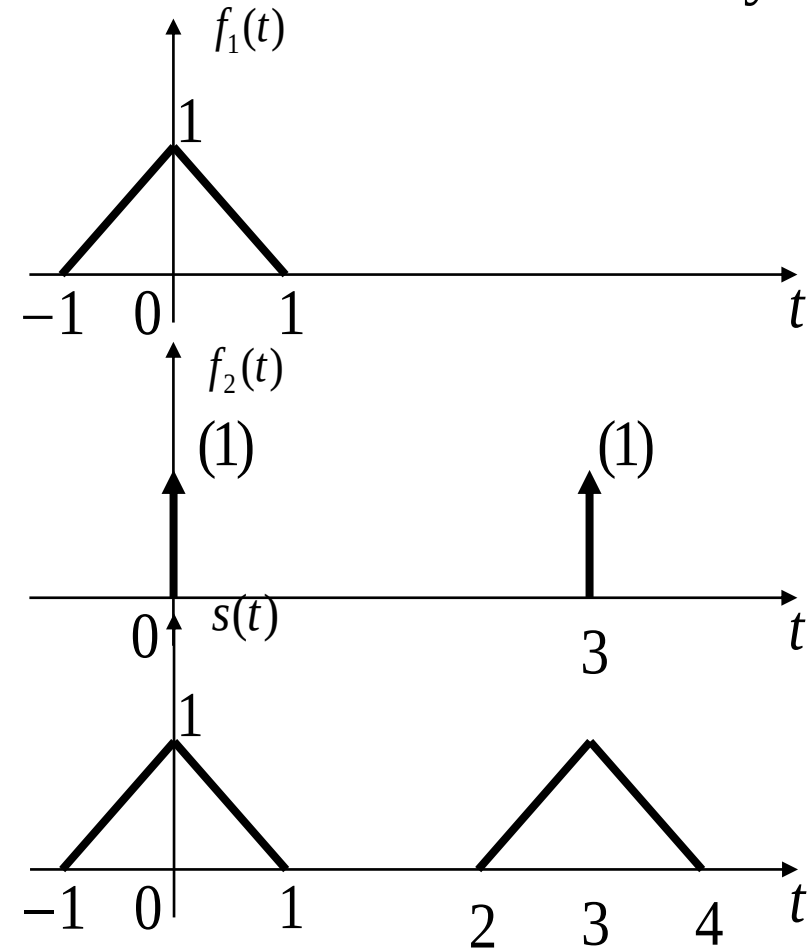
If $f(t) = f_1(t) * f_2(t)$, then $f_1(t-t_1) * f_2(t-t_2) = f_1(t-t_2) * f_2(t-t_1) = f(t-t_1-t_2)$

Physical Meaning: Shifting the excitation by t_1 and the impulse response by t_2 shifts the zero-state response by $t_1 + t_2$ —this reflects the **time-invariance** of LTI systems.

Example 2-11: Given the waveforms of $f_1(t)$ and $f_2(t)$, Find $s(t)=f_1(t)*f_2(t)$ and draw its waveform.

Solution: $f_2(t) = [\delta(t)+\delta(t-3)]$, then

$$\begin{aligned} s(t) &= f_1(t) * [\delta(t) + \delta(t-3)] \\ &= f_1(t) * \delta(t) + f_1(t) * \delta(t-3) \\ &= f_1(t) + f_1(t-3) \end{aligned}$$



6. Time Range of Convolution Integral

Example 2-12: Given that $f_1(t)$ has non-zero values in the interval $t_1 \leq t \leq t_2$ and $f_2(t)$ has non-zero values in $t_3 \leq t \leq t_4$. Let $f(t) = f_1(t) * f_2(t)$. Find the time range where $f(t)$ takes non-zero values.

Solution:

$$f(t) = f_1(t) * f_2(t) = \int_{-\infty}^{\infty} f_1(\tau) f_2(t - \tau) d\tau$$

The integral limits are determined by the **time overlap** between $f_1(\tau)$ and $f_2(t - \tau)$

$$t_1 \leq \tau \leq t_2,$$

$$t_3 \leq t - \tau \leq t_4 \rightarrow t - t_4 \leq \tau \leq t - t_3$$

If $t - t_4 > t_2$ (i.e., $t > t_2 + t_4$), or $t - t_3 < t_1$ (i.e., $t < t_1 + t_3$), there is **no time overlap** between $f_1(\tau)$ and $f_2(t - \tau)$, so the integral equals 0.

Conversely, $f(t)$ has non-zero values in the time range: $t_1 + t_3 \leq t \leq t_2 + t_4$.

The length of this interval is $(t_2 - t_1) + (t_4 - t_3)$, which is the **sum of the lengths of the non-zero intervals** of $f_1(t)$ and $f_2(t)$.

7. Differentiation Property

$$\frac{d}{dt}[f_1(t) * f_2(t)] = f_1(t) * \frac{df_2(t)}{dt} = \frac{df_1(t)}{dt} * f_2(t)$$

Physical Meaning: The derivative of the system's zero-state response equals the response to the derivative of the excitation (or vice versa). This property simplifies the analysis of systems with derivative-based excitations.

8. Integration Property

$$\int_{-\infty}^t [f_1(\lambda) * f_2(\lambda)] d\lambda = f_1(t) * \int_{-\infty}^t f_2(\lambda) d\lambda = \int_{-\infty}^t f_1(\lambda) d\lambda * f_2(t)$$

$$f_1(t) * f_2(t) = \frac{df_1(t)}{dt} * \int_{-\infty}^t f_2(\lambda) d\lambda = \int_{-\infty}^t f_1(\lambda) d\lambda * \frac{df_2(t)}{dt} \quad (f_1(t) \text{ and } f_2(t) \text{ should be 0 when } t \rightarrow -\infty.)$$

Physical Meaning: The integral of the zero-state response equals the response to the integral of the excitation—this complements the differentiation property and is useful for step responses (derived from impulse response integration).

2.8 Properties of Convolution



Commonly used convolutions

$$(1) K * f(t) = K \cdot [\text{area of } f(t)]$$

$$(2) f(t) * \delta(t) = f(t)$$

$$(3) f(t) * \delta'(t) = f'(t)$$

$$(4) f(t) * u(t) = \int_{-\infty}^t f(\tau) d\tau$$

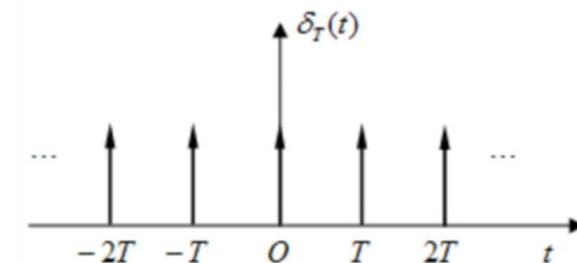
$$(5) u(t) * u(t) = tu(t)$$

$$(6) u(t) * e^{at}u(t) = -\frac{1}{a}[1 - e^{at}]u(t)$$

$$(7) e^{at}u(t) * e^{at}u(t) = te^{at}u(t)$$

$$(8) e^{a_1 t}u(t) * e^{a_2 t}u(t) = \frac{1}{a_1 - a_2} [e^{a_1 t} - e^{a_2 t}]u(t) \quad a_1 \neq a_2$$

$$(9) f(t) * \delta_T(t) = f(t) * \sum_{m=-\infty}^{+\infty} \delta(t - mT) = \sum_{m=-\infty}^{+\infty} f(t - mT)$$



- Repeat the examples in the lecture notes.
- 2.39(a)(b) and 2.48 in the textbook.